MPG No. 7.5.124 $m'm2'$ (mm'2' setting) [Type III, orthorhombic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} M_x & 0 & 0 \end{bmatrix}$$

$$M_x = \mathbb{M}_x^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & M_{xz} \\ 0 & 0 & 0 \\ M_{xz} & 0 & 0 \end{bmatrix}$$

$$M_{xz} = \mathbb{M}_{xz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & M_{xz} \\ 0 & 0 & 0 \\ -M_{xz} & 0 & 0 \end{bmatrix}$$

$$M_{xz} = -\mathbb{T}_y^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & 0 \\ M_{yyx} & 0 & 0 \\ M_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{zzx} \\ 0 & M_{yyx} & 0 \end{bmatrix}$$

$$M_{xxx} = 3\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)}$$
$$M_{yyx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\beta(1)}$$
$$M_{zzx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\beta(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ M_{yyx} & 0 & 0 \\ M_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{M_{zzx}}{2} \\ 0 & -\frac{M_{yyx}}{2} & 0 \end{bmatrix}$$

$$M_{yyx} = 2\mathbb{M}_x^{(2)} + 2\mathbb{T}_{yz}^{(1)}$$
$$M_{zzx} = 2\mathbb{M}_x^{(2)} - 2\mathbb{T}_{yz}^{(1)}$$
$$\frac{M_{zzx}}{2} = \mathbb{M}_x^{(2)} - \mathbb{T}_{yz}^{(1)}$$
$$\frac{M_{yyx}}{2} = \mathbb{M}_x^{(2)} + \mathbb{T}_{yz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{zxz} \\ 0 & M_{xyy} & 0 \end{bmatrix}$$

$$M_{zxz} = \mathbb{M}^{(3)}_{x} + \mathbb{T}^{(2)}_{yz}$$
$$M_{xyy} = -\mathbb{M}^{(3)}_{x} + \mathbb{T}^{(2)}_{yz}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ M_{xxzx} & M_{yyzx} & M_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = -\mathbb{M}^{\alpha(1)}_{4y} - \mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + 2\mathbb{M}^{(2)}_{xz}$$
$$M_{yyzx} = 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz}$$
$$M_{zzzx} = \mathbb{M}^{\alpha(1)}_{4y} - \mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + 2\mathbb{M}^{(2)}_{xz}$$
$$M_{yzxy} = 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(2)}_{xz}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ -M_{xxzx} & -M_{yyzx} & -M_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = \mathbb{M}^{(3)}_{xz} - 2\mathbb{T}^{(1)}_{y} + 2\mathbb{T}^{\alpha(1)}_{y} + 2\mathbb{T}^{\beta(1)}_{y}$$
$$M_{yyzx} = \mathbb{M}^{(3)}_{xz} - 4\mathbb{T}^{\beta(1)}_{y}$$
$$M_{zzzx} = \mathbb{M}^{(3)}_{xz} + 2\mathbb{T}^{(1)}_{y} - 2\mathbb{T}^{\alpha(1)}_{y} + 2\mathbb{T}^{\beta(1)}_{y}$$
$$M_{yzxy} = \mathbb{T}^{(1)}_{y} + 4\mathbb{T}^{\alpha(1)}_{y}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = 2\mathbb{M}^{(4)}_{xz}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = \mathbb{T}^{(2)}_{y}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & 0 \\ 0 & M_{yyzx} & 0 \\ 0 & M_{zzzx} & 0 \\ 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ M_{xyyz} & 0 & 0 \end{bmatrix}$$

$$
\begin{aligned}
M_{xxzx} &= 2\mathbb{M}^{(5)}_{xz} + \mathbb{T}^{(3)}_{y} - \mathbb{T}^{\alpha(2)}_{y} - \mathbb{T}^{\beta(2)}_{y} \\
M_{yyzx} &= \mathbb{T}^{(3)}_{y} + 2\mathbb{T}^{(4)}_{y} + 2\mathbb{T}^{\alpha(2)}_{y} \\
M_{zzzx} &= -2\mathbb{M}^{(5)}_{xz} + \mathbb{T}^{(3)}_{y} - \mathbb{T}^{\alpha(2)}_{y} + \mathbb{T}^{\beta(2)}_{y} \\
M_{yzxy} &= \mathbb{M}^{(5)}_{xz} + \mathbb{T}^{(4)}_{y} - \mathbb{T}^{\alpha(2)}_{y} + \mathbb{T}^{\beta(2)}_{y} \\
M_{xyyz} &= -\mathbb{M}^{(5)}_{xz} + \mathbb{T}^{(4)}_{y} - \mathbb{T}^{\alpha(2)}_{y} - \mathbb{T}^{\beta(2)}_{y}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & M_{yzxy} \\
M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & 0 & 0 \\
0 & 0 & 0 & M_{xyyz} & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{yzxy} &= -\mathbb{M}^{(6)}_{xz} + \mathbb{T}^{(6)}_{y} - \mathbb{T}^{\alpha(3)}_{y} - \mathbb{T}^{\beta(3)}_{y} \\
M_{zxxx} &= 2\mathbb{M}^{(6)}_{xz} + \mathbb{T}^{(5)}_{y} - \mathbb{T}^{\alpha(3)}_{y} - \mathbb{T}^{\beta(3)}_{y} \\
M_{zxyy} &= \mathbb{T}^{(5)}_{y} + 2\mathbb{T}^{(6)}_{y} + 2\mathbb{T}^{\alpha(3)}_{y} \\
M_{zxzz} &= -2\mathbb{M}^{(6)}_{xz} + \mathbb{T}^{(5)}_{y} - \mathbb{T}^{\alpha(3)}_{y} + \mathbb{T}^{\beta(3)}_{y} \\
M_{xyyz} &= \mathbb{M}^{(6)}_{xz} + \mathbb{T}^{(6)}_{y} - \mathbb{T}^{\alpha(3)}_{y} + \mathbb{T}^{\beta(3)}_{y}
\end{aligned}
$$

* Rank 4 tensor (s).

$$
\begin{bmatrix}
0 & 0 & 0 & 0 & M_{xxzx} & 0 & 0 & M_{xxxz} & 0 \\
0 & 0 & 0 & 0 & M_{yyzx} & 0 & 0 & M_{yyxz} & 0 \\
0 & 0 & 0 & 0 & M_{zzzx} & 0 & 0 & M_{zzxz} & 0 \\
0 & 0 & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzyx} \\
M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & M_{xyyz} & 0 & 0 & M_{xyzy} & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{xxzx} &= -\mathbb{M}^{\alpha(1)}_{4y} - \mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + 2\mathbb{M}^{(2)}_{xz} + \mathbb{M}^{(3)}_{xz} + 2\mathbb{M}^{(5)}_{xz} - 2\mathbb{T}^{(1)}_{y} + \mathbb{T}^{(3)}_{y} + 2\mathbb{T}^{\alpha(1)}_{y} - \mathbb{T}^{\alpha(2)}_{y} + 2\mathbb{T}^{\beta(1)}_{y} - \mathbb{T}^{\beta(2)}_{y} \\
M_{xxxz} &= -\mathbb{M}^{\alpha(1)}_{4y} - \mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + 2\mathbb{M}^{(2)}_{xz} + \mathbb{M}^{(3)}_{xz} - 2\mathbb{M}^{(5)}_{xz} - 2\mathbb{T}^{(1)}_{y} - \mathbb{T}^{(3)}_{y} + 2\mathbb{T}^{\alpha(1)}_{y} + \mathbb{T}^{\alpha(2)}_{y} + 2\mathbb{T}^{\beta(1)}_{y} + \mathbb{T}^{\beta(2)}_{y} \\
M_{yyzx} &= 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + \mathbb{M}^{(3)}_{xz} + \mathbb{T}^{(3)}_{y} + 2\mathbb{T}^{(4)}_{y} + 2\mathbb{T}^{\alpha(2)}_{y} - 4\mathbb{T}^{\beta(1)}_{y} \\
M_{yyxz} &= 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + \mathbb{M}^{(3)}_{xz} - \mathbb{T}^{(3)}_{y} - 2\mathbb{T}^{(4)}_{y} - 2\mathbb{T}^{\alpha(2)}_{y} - 4\mathbb{T}^{\beta(1)}_{y} \\
M_{zzzx} &= \mathbb{M}^{\alpha(1)}_{4y} - \mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + 2\mathbb{M}^{(2)}_{xz} + \mathbb{M}^{(3)}_{xz} - 2\mathbb{M}^{(5)}_{xz} + 2\mathbb{T}^{(1)}_{y} + \mathbb{T}^{(3)}_{y} - 2\mathbb{T}^{\alpha(1)}_{y} - \mathbb{T}^{\alpha(2)}_{y} + 2\mathbb{T}^{\beta(1)}_{y} + \mathbb{T}^{\beta(2)}_{y} \\
M_{zzxz} &= \mathbb{M}^{\alpha(1)}_{4y} - \mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + 2\mathbb{M}^{(2)}_{xz} + \mathbb{M}^{(3)}_{xz} + 2\mathbb{M}^{(5)}_{xz} + 2\mathbb{T}^{(1)}_{y} - \mathbb{T}^{(3)}_{y} - 2\mathbb{T}^{\alpha(1)}_{y} + \mathbb{T}^{\alpha(2)}_{y} + 2\mathbb{T}^{\beta(1)}_{y} - \mathbb{T}^{\beta(2)}_{y} \\
M_{yzxy} &= 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(2)}_{xz} + \mathbb{M}^{(5)}_{xz} + \mathbb{T}^{(1)}_{y} + \mathbb{T}^{(4)}_{y} + 4\mathbb{T}^{\alpha(1)}_{y} - \mathbb{T}^{\alpha(2)}_{y} + \mathbb{T}^{\beta(2)}_{y} \\
M_{yzyx} &= 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(2)}_{xz} - \mathbb{M}^{(5)}_{xz} + \mathbb{T}^{(1)}_{y} - \mathbb{T}^{(4)}_{y} + 4\mathbb{T}^{\alpha(1)}_{y} + \mathbb{T}^{\alpha(2)}_{y} - \mathbb{T}^{\beta(2)}_{y} \\
M_{zxxx} &= -\mathbb{M}^{\alpha(1)}_{4y} - \mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + 2\mathbb{M}^{(2)}_{xz} - \mathbb{M}^{(3)}_{xz} + 2\mathbb{T}^{(1)}_{y} - 2\mathbb{T}^{\alpha(1)}_{y} - 2\mathbb{T}^{\beta(1)}_{y} \\
M_{zxyy} &= 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} - \mathbb{M}^{(3)}_{xz} + 4\mathbb{T}^{\beta(1)}_{y} \\
M_{zxzz} &= \mathbb{M}^{\alpha(1)}_{4y} - \mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(1)}_{xz} + 2\mathbb{M}^{(2)}_{xz} - \mathbb{M}^{(3)}_{xz} - 2\mathbb{T}^{(1)}_{y} + 2\mathbb{T}^{\alpha(1)}_{y} - 2\mathbb{T}^{\beta(1)}_{y} \\
M_{xyyz} &= 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(2)}_{xz} - \mathbb{M}^{(5)}_{xz} - \mathbb{T}^{(1)}_{y} + \mathbb{T}^{(4)}_{y} - 4\mathbb{T}^{\alpha(1)}_{y} - \mathbb{T}^{\alpha(2)}_{y} - \mathbb{T}^{\beta(2)}_{y} \\
M_{xyzy} &= 2\mathbb{M}^{\beta(1)}_{4y} + \mathbb{M}^{(2)}_{xz} + \mathbb{M}^{(5)}_{xz} - \mathbb{T}^{(1)}_{y} - \mathbb{T}^{(4)}_{y} - 4\mathbb{T}^{\alpha(1)}_{y} + \mathbb{T}^{\alpha(2)}_{y} + \mathbb{T}^{\beta(2)}_{y}
\end{aligned}
$$

* Rank 4 tensor (a).

$$
\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzyx} \\
M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & M_{xyyz} & 0 & 0 & M_{xyzy} & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{yzxy} &= 2\mathbb{M}_{xz}^{(4)} - \mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(2)} + \mathbb{T}_{y}^{(6)} - \mathbb{T}_{y}^{\alpha(3)} - \mathbb{T}_{y}^{\beta(3)} \\
M_{yzyx} &= -2\mathbb{M}_{xz}^{(4)} - \mathbb{M}_{xz}^{(6)} - \mathbb{T}_{y}^{(2)} + \mathbb{T}_{y}^{(6)} - \mathbb{T}_{y}^{\alpha(3)} - \mathbb{T}_{y}^{\beta(3)} \\
M_{zxxx} &= 2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(5)} - \mathbb{T}_{y}^{\alpha(3)} - \mathbb{T}_{y}^{\beta(3)} \\
M_{zxyy} &= \mathbb{T}_{y}^{(5)} + 2\mathbb{T}_{y}^{(6)} + 2\mathbb{T}_{y}^{\alpha(3)} \\
M_{zxzz} &= -2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(5)} - \mathbb{T}_{y}^{\alpha(3)} + \mathbb{T}_{y}^{\beta(3)} \\
M_{xyyz} &= 2\mathbb{M}_{xz}^{(4)} + \mathbb{M}_{xz}^{(6)} - \mathbb{T}_{y}^{(2)} + \mathbb{T}_{y}^{(6)} - \mathbb{T}_{y}^{\alpha(3)} + \mathbb{T}_{y}^{\beta(3)} \\
M_{xyzy} &= -2\mathbb{M}_{xz}^{(4)} + \mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(2)} + \mathbb{T}_{y}^{(6)} - \mathbb{T}_{y}^{\alpha(3)} + \mathbb{T}_{y}^{\beta(3)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
0 & 0 & M_{xxxz} \\
0 & 0 & 0 \\
M_{zzzx} & 0 & 0 \\
M_{yyzx} & 0 & 0 \\
0 & 0 & M_{zzxz} \\
0 & 0 & 0 \\
0 & 0 & 0 \\
M_{zxxx} & 0 & 0 \\
0 & 0 & M_{xyyz} \\
0 & M_{xyyz} + \frac{M_{zzxz}}{2} - \frac{M_{zzzx}}{2} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{xxxz} &= -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} \\
M_{zzzx} &= \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} \\
M_{yyzx} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} \\
M_{zzxz} &= \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} \\
M_{zxxx} &= -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} \\
M_{xyyz} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)}
\end{aligned}
$$